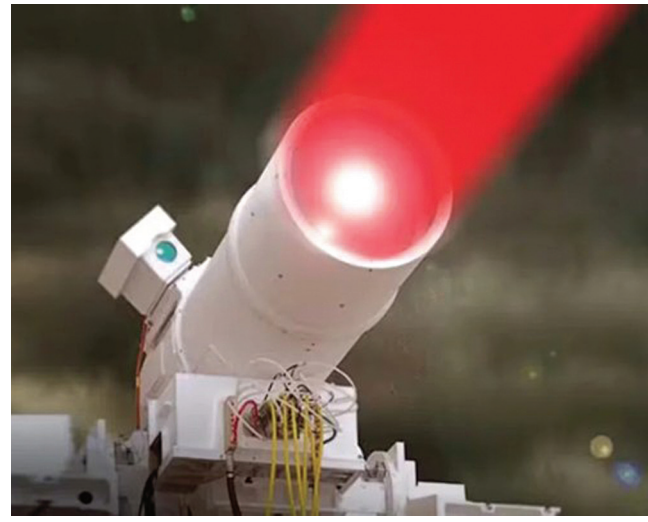
burden of transporting and storing massive quantities of ammunition. DEWs provide advanced electronic warfare capabilities, such as jamming to disrupt communication between drones and their operators or dazzling the sensors of spy satellites, temporarily blinding them. In space, DEWs provide a significant advantage, enhancing space situational awareness (SSA) (Box 2). Beyond 'laser brooming', which clears hazardous orbital debris, ground-based lasers safeguard sensitive assets by temporarily blinding or obscuring overhead surveillance systems. For public safety and crowd management, technologies like the active denial system (ADS) provide a critical, non-lethal intermediary 'middle ground' between verbal warnings and lethal force.

DEWs: India's initiatives

India has initiated a programme to develop DEWs as an integral part of Mission Sudarshan Chakra, a nationwide integrated shield to counter threats like drone swarms, precision-guided cruise missiles, and hypersonic missiles. The programme, highly classified in nature, is led by the DRDO. The leading projects include:



A single shot pulsed gigawatt electron accelerator – Part of the KALI weapon series (Source: https://zeenews.india.com)

DURGA-II Programme. The flagship of the programme, DURGA-II (Directionally Unrestricted Ray-Gun Array), is developed by the Laser Science and Technology Centre (LASTEC), a DRDO laboratory. DURGA-II is a 100-kilowatt-class lightweight system designed for 'hard-kill' maritime and aerial defence. While the 25kW-class systems are soft-kill 'dazzlers', DURGA-II fires a concentrated thermal beam travelling at the speed of light to physically combust drone structures, shatter artillery shells, and disable fast-moving missile guidance packages. Engineered as a modular architecture, the system can be seamlessly integrated across multi-domain platforms. DURGA-II is undergoing Indian Navy trials to evaluate harbour defence and anti-drone capabilities.

KALI project. Complementing DURGA-II is KALI (Kilo-ampere linear injector) developed jointly by DRDO and the Bhabha Atomic Research Centre (BARC). KALI is a linear electron accelerator that emits intense electron pulses. The pulses are converted into high-power microwaves to cause a 'soft-kill'. Also, KALI-5000 and KALI-10000 are two of the projects that are under development. KALI-5000 generates pulses reaching 40 gigawatts of power. KALI-10000 is primarily for research and also for testing the vulnerability of satellites and aircraft.

1GW high-power microwave (HPM) DEW. The 1GW HPM DEW has been developed by the Microwave Tube Research and Development

Box 2: Space situational awareness (SSA)

SSA involves monitoring, tracking, and predicting the movement of objects orbiting the Earth. As the orbits get increasingly crowded with satellites and fragments of space junk, SSA, as the 'traffic control system' for the cosmos, keeps a constant watch on the orbital highway to prevent catastrophic collisions and secure the global communication infrastructure, GPS navigation, and defence networks, keeping space safe, predictable, and sustainable.

TABLE 2
TECHNICAL CHARACTERISTICS OF DIRECTED ENERGY WEAPONS

DEW Type	Mechanism	Typical Power Levels	Effective Range	Applications	Examples/Status
High-energy lasers	Coherent light beam heats/damages target	10kW-100kW (prototypes)	1km-5km (current systems)	Neutralising drones, missiles, and sensors	DURGA-II (100kW, in development), Vehicle-Mounted MK-II, H.A.R.A. MK-1 (10kW)
High-power microwaves	Electromagnetic pulses disrupt electronics	Tens of MW (pulse power)	Short to medium range (hundreds of metres to a few kms)	Disabling UAV swarms, radar, guidance systems	CHESS prototypes (short-range UAV defence)
Particle beam weapons	Accelerated charged particles cause damage at the molecular level	Experimental, very high power	Limited, lab-scale only	Potential against hardened targets, satellites	Only at research stage, no prototypes yet
Acoustic/ Hybrid	Directed sound or pressure waves	kW-level (non-lethal)	Short range (<100m)	Crowd control, disabling sensitive electronics	Non-lethal acoustic devices under study